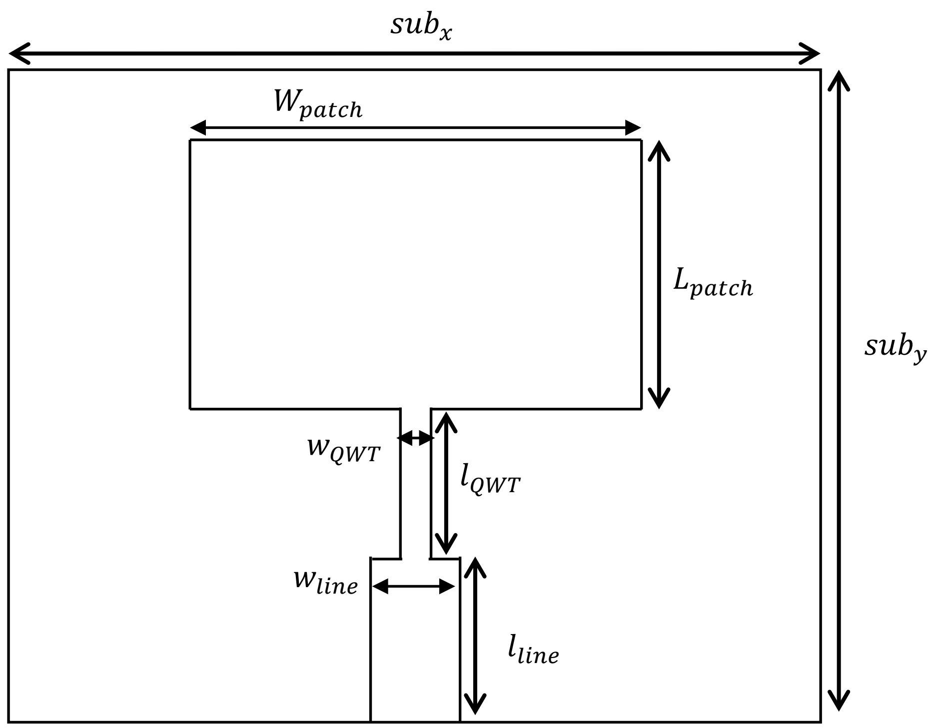
Lab. 3 – Numerical and experimental analysis of Patch Antennas

Assigned: **Friday April 24, 2026**
Due: **Friday May 8, 2026**

Notes
To take the exam with the grading of submitted assignments + oral discussion it is necessary:
- to prepare **individually** the report of the lab, that must include the Matlab script, the obtained results and their brief comment;
- to submit the report **within the deadline**, uploading the PDF in the section "Elaborati" of the web portal

The lab is focused on the numerical and experimental characterization of the rectangular patch and the feeding line shown in Figure. They are printed on a FR4 substrate, with relative dielectric constant ε_r= 4.45 and thickness h = 1.55mm. The central frequency is f_0 = 5.8 GHz.



PART 1

Analyze the antenna using CST Studio Suite in the two following cases:

- ideal substrate → $\boldsymbol{tan\delta = 0}$
- real substrate → $\boldsymbol{tan\delta = 0.0188}$

In both the cases:

- evaluate the frequency behavior of the input scattering parameter S_{11} of the antenna over the interval [5.3 – 6.3] GHz and plot the behavior of $|S_{11}|$ (in dB);
- evaluate and plot the radiation pattern at the central frequency in the E and H planes; determine the polarization of the patch and the value of the (max) gain in the direction of maximum radiation; compare with the plots obtained in the Problem 2 of assignment 3, paying attention to the different plane in which the patch might be located.

These are the geometrical parameters for the CST Studio Suite project:

- $sub_x = 50\ mm$
- $sub_y = 45\ mm$
- $W_{patch} = 32\ mm$
- $L_{patch} = 10.65\ mm$
- $w_{QWT} = 1.3\ mm$
- $\ell_{QWT} = 7.2\ mm$
- $w_{line} = 2.7\ mm$
- $\ell_{line} = 17\ mm$

PART 2

Using the VNA (after its calibration), measure the variation of $|S_{11}|$ of the antenna prototype over the same frequency range considered in PART 1.

Compare the results obtained by the experimental characterization with those deriving from its numerical simulation (use the substrate with losses). Represent the frequency behavior of $|S_{11}|$ (in dB) on the same plot and comment the obtained results.

Eventually, compare the behavior of this antenna with the one realized with insets, again showing the behavior of $|S_{11}|$ (in dB) on the same plot.